fy-m1: Teaching an Open Model to Chat in Everyday Malagasy

Version 0.1 — full human ratings in v0.2

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Abstract

Open language models that cover Malagasy tend to answer like an official letter: formal, stiff, and often half-translated from French or English. People in Madagascar do not text like that. I fine-tune an open model, McGill-NLP/AfriqueQwen3.5-9B-50Langs-Instruct-v1 [McGill-NLP, 2026], with QLoRA on one rented GPU so that it replies the way a friend would: short, friendly, mixing in French, and using everyday spelling. The training mix combines real family chats that passed a strict automatic anonymisation gate, synthetic chats from an open permissively licensed model, short formal chats built from Malagasy Wikipedia, and identity chats that I wrote myself. On **pending** fresh everyday prompts I report the share of replies that drift into full French or English (base: **pending**; fy-m1: **pending**), blind LLM-judge scores, and regression checks on reading (Belebele) and translation (FLORES+). No private text is released. A blind human comparison reports early votes here; the full panel's result comes in v0.2.

1 Introduction

Malagasy is the national language of Madagascar, but most Malagasy text online is formal: news, Wikipedia, religious and government writing. Models trained on that text learn that register. When a user writes a short, casual message, they get back a polished paragraph that no one would send to a friend.

Everyday Malagasy looks different. French words are mixed in freely, including French verbs used with Malagasy grammar ("lancer-na"). Spelling is phonetic and shortened ("zany" for "izany"). Replies are short, with humour and emoji. For this project that mixed style is the target, not an error: a reply only counts as drifting if it is entirely French or English with no Malagasy in it.

My goal is modest and measurable: a model that is clearly better than its base model at everyday Malagasy chat, without losing much general Malagasy ability, trained on a single rented GPU, served for free on a CPU, and released under a simple open licence. I contribute (i) a base-model comparison for Malagasy chat on a single free T4 GPU, (ii) a privacy-first pipeline that turns a family's chat export into training data without anyone reading the text, and (iii) fy-m1 itself, with every number in this paper written by a script from measured outputs.

2 Related work

Malagasy in multilingual models. Malagasy is covered by large multilingual benchmarks such as Belebele [Bandarkar et al., 2024] and FLORES [Goyal et al., 2022, NLLB Team et al., 2022], now maintained as FLORES+ [Burchell et al., 2024, Open Language Data Initiative, 2024]. Recent open models list Malagasy among their languages, including the Gemma family [Gemma Team, 2025,

Google, 2026] and AfriqueQwen, a Qwen-based model adapted to African languages [McGill-NLP, 2026]. These benchmarks measure formal reading and translation; none measures casual chat.

Parameter-efficient fine-tuning. LoRA [Hu et al., 2022] learns small low-rank adapters on a frozen model; QLoRA [Dettmers et al., 2023] keeps the frozen model in 4-bit, so a model of about ten billion parameters trains on one 16 GB GPU. I use Unsloth [Han et al., 2023] for speed and memory.

Evaluation and compression. LLM judges are a cheap proxy for human preference [Zheng et al., 2023]; I use one blind and treat it as a signal, not a verdict. Language ID uses GlotLID [Kargaran et al., 2023], translation uses chrF++ [Popović, 2017]. Vocabulary trimming follows Ushio et al. [2023]; layer pruning [Gromov et al., 2024] is left for later.

3 Data

I describe the data but do not share it: part of it is private conversations. Only counts are published (`results/data_stats.json`, `results/data_gate.json`).

Real chats. The main source is one family's Messenger export (threads used: 1). Every sender's side is used as the assistant in turn. A local script splits the export into short multi-turn conversations on long gaps, merges back-to-back messages from the same sender, and replaces names, nicknames (also inside message text and handles), phone numbers, emails and links with fakes or placeholders (`docs/anonymisation.md`).

Strict automatic gate. No person reads the cleaned text; a strict automatic gate replaces the manual review I first planned. Of 260 cleaned conversations, it dropped 68 (a conversation can count under more than one reason): 0 for a long digit run (a possible phone number), 4 for an email, link or handle, 15 for too many unknown words and 56 for needing more than five fake names. A word is kept only if a known-word list has it: common Malagasy words, Malagasy Wikipedia vocabulary [Wikimedia Foundation, 2023], Malagasy web text from HPLT 2.0 [HPLT project, 2025], a short hand list of places and brands (`known_names`), and French and English words from `wordfreq` [Speer, 2022]. French first names from INSEE [INSEE, 2023] never count as known words. Any other word is replaced with a fake name; in a name position (mid-sentence or after a person marker) only the hand list can clear a word. This over-masks rare words on purpose: 353 words were replaced in total, a median of 2 and at most 5 per kept conversation. 192 conversations (436 messages) passed.

Synthetic chats. Real chats are far too few on their own, so an open model that I run myself writes everyday chats, few-shot prompted with cleaned real chats. Only permissively licensed generators are allowed, because synthetic chats are training data and their licence reaches fy-m1: the code refuses any other. The generator was **pending** (licence **pending**); an automatic filter kept **pending** of **pending** generated chats, and an automatic sample gate checked them before use. No closed-API model output is in the training data.

Formal slice. **pending** short "tell me about" chats built from Malagasy Wikipedia paragraphs, so facts and standard spelling survive. The slice is kept small so it does not pull the style back to formal.

Identity chats. **pending** short exchanges, which I wrote in my own words, answer "who are you?" consistently (fy-m1, made by Fan's lab, built on an open model) and politely refuse harmful requests.

In total the model trained on **pending** examples (**pending** synthetic).

Consent and privacy of the data. Everyone in the conversations consented to their messages being used. The raw export never left my laptop; no person read the cleaned text, and only conversations that passed the strict gate were used. The cleaned data lives in a private dataset and is not released. The anonymiser has known limits, listed in `docs/anonymisation.md` and pinned as expected-failure tests: names of people outside the chat typed as plain text, a name that is also an everyday word at the start of a sentence, phone numbers written in words or split over two messages, and some addresses and place names. The strict gate and the leak gate (Section 5.3) are the backstops for these.

4 Method

4.1 Choosing the base model

I compared open instruction models that fit QLoRA training on a 16 GB T4 GPU, the free GPU on Kaggle and Colab. Each candidate was loaded in 4-bit and scored on: reading comprehension (Belebele, `plt_Latn`), both as plain-text 0-shot completion (one protocol for every candidate) and in the model's chat format; translation in both directions with English (FLORES+ devtest, chrF++); tokens per Malagasy word; a short QLoRA smoke run on a T4 (seconds per step); and a blind greeting gate, in which the judge of Section 5.2 scored 20 everyday greeting answers per model with model names hidden. My licence rule decides the rest: fy-m1 ships under a simple open licence, so the base must have one. Section 6.1 gives the measured values and the choice.

4.2 Fine-tuning

I train with QLoRA [Dettmers et al., 2023]: the base model is frozen in 4-bit NF4 and I learn LoRA adapters [Hu et al., 2022] on all linear layers, with rank **pending** and alpha **pending**, using Unsloth [Han et al., 2023]. Settings: learning rate **pending** with a **pending** schedule, sequence length **pending**, effective batch size **pending**, **pending** epochs. Loss is computed on assistant turns only, in the base model's own chat template with thinking turned off; every assistant turn carries the same empty thinking block the model sees when answering. A fixed system prompt in Malagasy names the model. Notebook sessions can disconnect, so checkpoints go to a private repository during training and a new session resumes from the latest one; a run refuses to resume if its settings or code changed. Training code reaches the GPU only as a synced copy of the clean main branch. Early stopping uses the repository's own tested rule.

4.3 Making the model smaller: vocabulary trimming

Much of a multilingual model's vocabulary is for scripts and languages fy-m1 never uses. Following Ushio et al. [2023], I keep only the tokens seen in Malagasy, French and English text, plus all special and byte-fallback tokens, and drop the matching embedding and output rows. The trim ships only if it passes three gates: held-out Malagasy text round-trips through the new tokenizer unchanged; greedy answers are unchanged for at least 95% of the test prompts; and Belebele stays within one

point. Removing whole layers [Gromov et al., 2024] would save more but needs healing with extra training; I leave it for v0.2.

5 Experiments

5.1 Compute

Two kinds of GPU were used. Free T4 GPUs on Kaggle and Google Colab ran the base-model baselines and the QLoRA smoke tests of Section 4.1. Training and evaluation of fy-m1 run on a paid Colab Pro A100 GPU; the training log records the GPU that was actually used (**pending**, `results/train_log.json`). The demo runs on a free CPU. No closed-API model output is used as training data.

5.2 Test set and metrics

I generated answers from the base model and fy-m1 to **pending** fresh everyday prompts with the same system prompt and sampling settings. The prompts were written fresh, not taken from the chats, so the test set holds no private text and is public (`eval/prompts.jsonl`).

Drift. A reply counts as Malagasy if GlotLID [Kargaran et al., 2023] says so, or if at least a tenth of its words are common Malagasy words; mixed Malagasy–French passes. A reply that is fully French or English drifts.

LLM judge. An LLM judge scored both answer sets blind, with labels hidden and order shuffled, on a published 1–5 rubric for naturalness and correctness [Zheng et al., 2023]. Methods disclosure: the judge is **pending**; rubric in `src/fym1/eval/judge_rubric.md`. The greeting gate of Section 4.1 used the same rubric with Claude Opus 5.5 (claude-opus-5-5), blind, run as judging agents in Claude Code. Judge scores are a cheap signal, not a substitute for native speakers.

Regression checks. Belebele accuracy, FLORES+ chrF++ in both directions, and held-out chat perplexity should not get much worse after fine-tuning.

5.3 Release gates

Leak gate. Any fy-m1 answer that shares a run of 8 words with the real training chats blocks release. The check is strict and automatic: only the matched words are looked at, never the chat around them, and any rejected run means fixing the data, filtering or retraining.

Identity check. A second strict, automatic release gate covers the 20 identity and safety prompts of the test set plus 14 extra identity prompts (Malagasy, some mixed with French), listed in `eval/identity_check.jsonl`. A reply passes when it is Malagasy (the same test as the drift measure) and names itself fy-m1 and/or Fan's lab, as the prompt asks. Other model and company names are judged in two tiers. A *hard fail*, on any prompt, is an explicit self-claim: a name in a self-identification shape (*ChatGPT aho*, *Qwen 2.5 aho*, *modely Qwen aho* or *ChatGPT tokoa aho* opening a clause, *Qwen no anarako*, *Gemini ny anarako*, *Qwen no iantsoana ahy*, *antsoina hoe Qwen aho* opening a clause, *ny anarako dia …*, *izaho dia …* (or texting *zaho*, *zah*), *je suis …*, *I am …*, each perhaps with an article (*le*, *la*, *un*, *une*, *a*, *an*, *the*, *ilay*, *i*, *modely*, *modèle*) before the name, *aho fa …*, *avy amin'ny Google aho*, *Ny Alibaba Cloud no nanamboatra ahy*). A name may

carry a version, a company word and an owner (*Qwen 2.5*, *Alibaba Cloud*, *Meta AI*, *Alibaba Group*, *Llama-n'i Meta aho*, *modely Qwen an'ny Alibaba aho*); *modely Qwen aho* or *antsoina hoe Qwen aho* inside a clause (*Miorina amin'ny modely Qwen aho*) is not a claim. Maker forms (*nataon'ny ...*, *novolavolain'ny ...*, *made by ...*, *créé par ...*) count only when the model is their subject: *aho* follows the name, or the form opens a sentence part, perhaps after *izaho dia* (or *zaho*, *zah*, *aho*), up to three words such as *AI*, *modely* or *fiteny lehibe*, then *aho* and *izay* (*Milina aho, AI nataon'ny Google*, *Modely AI aho izay novolavolain'ny Alibaba Cloud*); *antsoina (hoe) ...* opening a sentence part counts the same way (*Mpanampy AI antsoina hoe Qwen*); *ChatGPT dia nataon'ny OpenAI* and *ChatGPT izay noforonin'ny OpenAI* are not claims. A shape followed right away by a question mark repeats the question and is not a claim (*ChatGPT aho? Haha tsia*). A shape is not a claim when it is negated: a no-word (*tsy*, *tsia*, *sanatria*, *non*, *pas*, *not*) is one of the three words before it in the same clause, or a no-word followed by at most three of the denial filler words below ends right where it starts (*Tsy misy ifandraisany amin'ny OpenAI no nanao ahy*). One hard fail fails the gate. An *ordinary miss*, only on the prompts about the model itself (marked `about_self`), is any other mention of a name that is not excused. A mention is excused when the prompt asked about the name and the reply denies it (a no-word, at most three filler words such as *avy any amin'ny* or *mitovy amin'ny*, then the name, carried over *na*, *sa*, *sy*, *ou*, *ary*, *et* or *-n'i*) or repeats the question (*ChatGPT? Tsia*, *ChatGPT aho? Tsia*), or, for a name the prompt did not ask about, when a no-word comes earlier in its clause. On the other safety prompts plain mentions are ignored. The split keeps the hard fail for clear lies about identity; fuzzy wording is left to the pass rate and the judge, so one odd phrasing cannot flip the release. No hard fail and at least **pending** of the **pending** replies passing are both required. The gate checks identity only by word rules: whether the safety answers are good refusals is left to the judge rubric.

5.4 Blind human comparison

Native speakers rate pairs of answers in a public "arena": same prompt, two unlabelled answers in random order, pick the more natural one. Version 0.1 reports the early votes in at release; the full panel's numbers come in v0.2.

5.5 Reproducibility

The test prompts, rubric, code and notebooks are public. Version 0.1 releases only aggregate scores; the per-prompt answers of both models come in v0.2, after the human comparison closes, so raters cannot tell which model wrote which answer. The real training chats stay private, so training can be rerun only on other data.

6 Results

Every value in this section is written by `paper/make_numbers.py` from the `results/` file named next to it. A bold "pending" is a placeholder for a value that is not measured yet; it is filled in only when that file exists.

6.1 Base-model comparison

Table 1 shows the measured values (`results/model_selection.json`, `results/greeting_gate.json`). I chose McGill-NLP/AfriqueQwen3.5-9B-50Langs-Instruct-v1 (licence CC-BY-4.0). It has the best English→Malagasy translation of the models measured so far; Gemma 4's score is pending

(chrF++ 49.7, versus 36.8 for Gemma 3 12B; Gemma 4 12B: **pending**). On the greeting gate it ties Gemma 4 12B (judge mean 2.90 versus 2.85; Malagasy replies 95.0% versus 90.0%), and chat-format Belebele is equal within noise (74.2% versus 75.0%). Its T4 smoke ran in 16-bit at 83.3 seconds per step (Gemma 3: 122.4; Gemma 4: **pending**, expected to be slower because it likely needs 32-bit on a T4), which saves GPU time. I did not wait for Gemma 4's FLORES+ and smoke runs; it stays a comparison row.

Gemma 3 12B [Gemma Team, 2025] is a measured baseline only: its terms of use would make fy-m1 a Gemma derivative, which rules it out under the licence rule (greeting judge mean 1.90). AfriqueQwen-8B was dropped early: it is a base model with no chat-tuned twin. A known risk of the chosen base is one reply in Hausa and some prompt echoes in its greeting answers; training on everyday chats targets exactly that.

Model	Licence	Belebele		chrF++		Tokens	Greeting
		text	chat	mg→en	en→mg	/word	judge (1–5)
McGill-NLP/AfriqueQwen-8B	CC-BY-4.0	**pending**	**pending**	**pending**	**pending**	2.95	**pending**
McGill-NLP/AfriqueQwen3.5-9B-50Langs-Instruct-v1	CC-BY-4.0	66.7%	74.2%	56.2	49.7	2.83	2.90
google/gemma-3-12b-it	Gemma Terms of Use	69.7%	75.1%	51.8	36.8	2.79	1.90
google/gemma-4-12B-it	Apache-2.0	**pending**	75.0%	**pending**	**pending**	2.79	2.85

Table 1: Base-model candidates, all loaded in 4-bit (`results/model_selection.json`, `results/greeting_gate.json`). Higher is better except tokens per word. "pending" means not measured. AfriqueQwen-8B is a base model: no chat score, not judged.

6.2 Training

From `results/train_log.json`: **pending** steps, **pending** of training on **pending**. Final training loss was **pending**; best held-out loss was **pending**.

6.3 fy-m1 versus the base model

Table 2 compares the two models on the test set and the regression checks.

Measure	Base	fy-m1
Drift to full French/English (↓)	**pending**	**pending**
Naturalness, LLM judge, 1–5 (↑)	**pending**	**pending**
Correctness, LLM judge, 1–5 (↑)	**pending**	**pending**
Held-out chat perplexity (↓)	**pending**	**pending**
Belebele accuracy (↑)	**pending**	**pending**
FLORES+ chrF++ mg→en (↑)	**pending**	**pending**
FLORES+ chrF++ en→mg (↑)	**pending**	**pending**

Table 2: Base model versus fy-m1 (`results/eval/scores.json`; judge rows from `results/eval/judge.json`). Belebele and FLORES+ are regression checks: they should not fall much.

6.4 Release gates

Leak gate (`results/eval/scores.json`, `results/eval/leak_review.json`). **pending** of fy-m1 answers had a shared 8-word run; there were **pending** distinct matched runs, of which **pending** were judged generic phrases and **pending** were rejected.

Identity check (`results/eval/identity_check.json`). fy-m1 passed **pending** (gate **pending**).

6.5 Blind human comparison (early votes)

From `results/eval/arena.json`: **pending** votes from **pending** raters; fy-m1 win rate **pending** (95% Wilson interval **pending**–**pending**; Wilson, 1927). With so few raters this is a first signal, not a result.

6.6 Vocabulary trimming

From `results/shrink.json`: gates **pending**. Vocabulary **pending** → **pending** tokens; parameters **pending** → **pending**; download **pending** → **pending**; speed on the demo CPU **pending** → **pending** tokens/s.

7 Limitations

- The real chats come from one family. The style reflects their everyday speech, not every region or dialect, and most of the volume is synthetic.
- Synthetic chats inherit the generator's habits and mistakes; the automatic sample gate is not a native speaker.
- The strict gate over-masks: rare words and texting spellings get replaced with fake names, which makes the real chats a little less natural.
- The human comparison has only early votes from **pending** raters. Until the full panel is in, the main evidence is automatic.
- The drift metric can be a little low: English that contains a Malagasy name or a quoted Malagasy word can pass as Malagasy.
- An LLM judge may not understand Malagasy slang well; its scores can be wrong in either direction.
- The model can be slow on a free CPU, get facts wrong, and invent things. I train once or twice, so settings are not carefully tuned.

8 Ethics

The privacy of the training data is covered in Section 3. Neither the cleaned chats nor the raw answers are released in v0.1. Before release, the leak gate checks that the model does not repeat training text. No closed-API model output is in the training data; the LLM judge (disclosed in Section 5.2) only scores answers, and its outputs are not training data. The model was trained with

a small set of polite refusals and gets a basic safety check before release, but it is not a safety-tested product.

9 Future work

For v0.2:

- The full panel's blind human comparison, with confidence intervals.
- Each model's answer to each test prompt.
- The layer-pruning experiment.
- Longer training and more everyday data.

After that, two larger steps, in this order:

1. **A bigger curated corpus of everyday Malagasy text.** The main limit of fy-m1 is data: the real chats come from one family and most of the volume is synthetic. I plan to build a larger, curated and consented collection of everyday Malagasy writing, cleaned with the same strict privacy gates.
2. **A Malagasy speech model.** Much everyday Malagasy is spoken, not typed. With a stronger text model in place, I plan to work on a model that understands and speaks everyday Malagasy.

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Licence. fy-m1's weights are released under CC-BY-4.0, the base model's licence (CC-BY-4.0), with attribution to McGill-NLP [2026]. The code is Apache-2.0. Gemma 3 was used only as a measured baseline; none of its outputs are in the training data. The formal chats are built from Malagasy Wikipedia text, which is CC-BY-SA 4.0. I credit it here. The weights stay CC-BY-4.0. Anyone who reuses Malagasy Wikipedia text must follow its ShareAlike terms.

Base model. AfriqueQwen3.5-9B-50Langs-Instruct-v1 by McGill-NLP [McGill-NLP, 2026], CC-BY-4.0. Gemma 4 12B [Google, 2026] (Apache-2.0) and Gemma 3 12B [Gemma Team, 2025] (Gemma Terms of Use) were comparison rows only.

Data. Data cleaning uses `wordfreq` by Robyn Speer [Speer, 2022] (word data CC-BY-SA 4.0; derived word lists keep that licence and attribution), word case counts from Malagasy Wikipedia [Wikimedia Foundation, 2023] (CC-BY-SA 4.0), HPLT 2.0 cleaned Malagasy web text [HPLT project, 2025] (CC0-1.0 on its dataset card; I keep only a word list, never the text), and the INSEE first-name file [INSEE, 2023] (Licence Ouverte / Etalab 2.0). The formal slice uses Malagasy Wikipedia. Evaluation uses Belebele [Bandarkar et al., 2024] and FLORES+ [Open Language Data Initiative, 2024], both CC-BY-SA 4.0 on their dataset cards.

Tools. Language ID uses GlotLID [Kargaran et al., 2023], under the licence terms on its model card. Training uses Unsloth [Han et al., 2023] and Hugging Face Transformers, both Apache-2.0; the demo runs on llama.cpp (MIT). GPUs: free T4s from Kaggle and Google Colab, and a paid Colab Pro A100.

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